

Post-Quantum Traceable Anonymous Cross-Domain Authentication for Blockchain-Based IoT

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Abstract—With the proliferation of Internet of Things (IoT) applications, secure and efficient cross-domain information sharing has become essential. Traditional cross-domain authentication approaches that rely on real device identities pose significant privacy risks, while fully anonymous schemes often lack accountability for malicious behavior. Additionally, emerging quantum computing capabilities threaten the security of classical cryptographic systems. To address these challenges, this paper proposes a post-quantum secure, traceable anonymous cross-domain authentication scheme based on blockchain technology. The scheme incorporates lattice-based certificateless cryptography and quantum-resistant hash functions to ensure resilience against quantum attacks. Devices are assigned multiple unlinkable pseudonymous identities with corresponding post-quantum public-private key pairs. A dynamic accumulator is employed to update domain information, and distinct pseudonyms are used for each authentication. Cross-domain credentials issued by a key generation center enable identity verification without exposing real identities. The system ensures both device privacy and traceability of malicious actors. Formal security proofs and BAN logic analysis confirm the scheme's robustness against classical and quantum-era threats. Compared with existing classical approaches, the proposed scheme achieves strong post-quantum security with reduced computational and communication overhead during the authentication process.

Index Terms—Post-quantum cryptography, cross-domain authentication, blockchain, traceable anonymity, Internet of Things, certificateless cryptography, lattice-based cryptography.

I. INTRODUCTION

THE Internet of Things (IoT) [1] connects smart devices such as sensors and processors to form vast, heterogeneous networks that generate and exchange real-time data. IoT is applied across domains—smart homes, healthcare, industrial automation, and vehicular networks—enhancing efficiency and quality of life. With an expected surge to 75 billion devices by 2025 [2], security and privacy challenges are increasingly critical. For instance, wearable devices collecting health data or vehicular networks tracking location

can expose users to privacy breaches if identity protection is absent [3], [4].

A Post-Quantum Traceable Anonymous Cross-Domain Authentication for IoT based on Blockchain leverages blockchain for decentralized trust and traceability while securing IoT communications against future quantum attacks using sophisticated post-quantum cryptographic algorithms. The main purpose of a transitional hybrid post-quantum (PQ) solution is to maintain security and backward compatibility while pre-quantum techniques are abandoned by fusing new PQ algorithms with classical encryption. A fully PQ method, on the other hand, only uses quantum-resistant algorithms and completely eliminates the older cryptographic elements. system for authentication that uses blockchain technology. Lattice-based certificateless cryptography and quantum-resistant hash functions are incorporated into the scheme to guarantee resiliency against quantum assaults. With features like user anonymity and unlinkability, such a system ensures data integrity and privacy even as the network expands by allowing devices from various domains to safely authenticate one another. To safeguard local resources, organizations deploy independent identity authentication systems, creating isolated trust domains [5]. Yet, many IoT services require collaboration across these domains [6]. Due to concerns over foreign entities, cross-domain cooperation remains limited unless a secure and privacy-preserving authentication mechanism is in place [7].

Cross-domain authentication faces challenges of mutual distrust, dynamic participation, and the need for decentralized trust management. Blockchain offers a promising solution by leveraging decentralization, immutability, and traceability [8]. Consortium blockchains, in particular, restrict node participation via authorization, making them ideal for regulated, multi-domain collaboration [4], [9], [10]. Techniques such as idempotency, retries with backoff, idempotent message queues, eventual consistency patterns, and optimistic concurrency management are necessary for managing accumulator updates in unstable networks. Retries are used to manage delays, reattempts and eventual data convergence are allowed to solve intermittent connectivity, and robust error detection, fallback methods, and redundancy are used to handle partial failures. Make certain that an operation can be carried out more than once without affecting the outcome. In the event that a message is transmitted more than once as a result of network retries, this stops redundant updates from polluting the accumulator.

However, protecting device identity while allowing malicious behavior to be traced is a difficult tradeoff. If pseudonyms are reused, attackers can link sessions and infer

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user behavior. Conversely, strong anonymity can also shield malicious actors. A balanced design must ensure privacy, accountability, and traceability [11]. Formal security analysis validates the scheme's resistance to typical assaults, including those that could occur in a post-quantum setting, and BAN logic is used to verify correctness. The scheme's low computational and communication overhead, as demonstrated by performance evaluation, makes it appropriate for IoT contexts with constraints. Understanding latency's components, assessing its effects, and then optimizing your system in light of the analysis are all necessary to improve communication overhead analysis with network latency. Data transfer delays, or network latency, are a crucial consideration, particularly for real-time applications where lag can impair user experience and performance. Strong malicious device detection and accountability are established by the suggested method, which promotes safe multi-domain collaboration by striking a balance between anonymity and traceability.

Anonymous authentication techniques [12], [13] commonly employ pseudonyms. Yet, schemes such as those in [14] and [15] suffer from pseudonym reuse, allowing session linking and privacy leakage. Improvements like those by Zeng et al. [16] and Feng et al. [17] introduce pseudonym updates but still lack traceability. Others, like Ali et al. [18], Zhao et al. [19], and Chen et al. [20], support identity recovery via trusted authorities, yet they still risk linkability due to pseudonym validity windows [21].

Post-quantum protocols come with a wealth of primitives that can be applied to a variety of use cases and to address implementation issues across different computing platforms (such as cloud versus IoT ecosystems) [22]. Based on asymmetric key cryptography (using quantum machines and algorithms), this entails the capacity to calculate encrypted data by having robust protocols against strong attackers and to offer security outside of the realm of traditional cryptography [23]. The protection of mission-critical infrastructures, particularly in the fields of energy, medicine, surveillance, space exploration, etc., is something that PQ cryptosystems are dedicated to enhancing [24]. PQ cryptosystems' scalability and versatility make them suitable for use in next-generation 5G/NB-IoT networks, as well as for secure communications in infrastructure related to electric vehicle charging [25].

In addition, the emergence of quantum computing threatens traditional cryptographic assumptions. Current cross-domain authentication mechanisms must evolve to incorporate post-quantum security primitives that can resist quantum attacks while maintaining lightweight performance on constrained IoT devices. The assertion of a quantum-resistant scheme's security against quantum attacks should be reinforced by addressing the hardness assumptions of the underlying LWE problem explicitly and citing recent cryptanalysis results that support this assumption. For instance, pointing out that the system makes use of a standardized algorithm such as Module-LWE-based CRYSTALS-Kyber lends legitimacy to its quantum resistance. It would also be helpful to explain how the security of the system is preserved in spite of developments in quantum algorithms.

To address these challenges, this paper proposes a blockchain-based, traceable, and anonymous cross-domain authentication scheme tailored for IoT environments. The scheme leverages post-quantum cryptography and blockchain features to ensure device privacy, accountability, and resilience against quantum-capable adversaries. Using a multi-layered strategy of hardware, software, and protocol-level defenses, side-channel attack resilience reduces information leakage through physical channels such as timing, power consumption, and electromagnetic radiation. Hardware shielding and filtering, constant-time and masking implementations, and secure protocols such as the use of Hardware Security Modules (HSMs) are important tactics. To prepare for, resist, and recover from attacks, a complete strategy integrates several techniques. Public-private key pairs are created using certificateless cryptography to improve privacy and remove the possibility of private key leaking from PK key escrow. By storing only the total value of each domain on the blockchain in a dynamic accumulator, the approach preserves authentication functionality while lowering communication and storage overhead that would result from uploading all pseudonymous public keys to the blockchain. Our main contributions are as follows:

- 1) We design a secure and efficient cross-domain authentication protocol based on a consortium blockchain. It supports bidirectional authentication and key agreement while employing a dynamic accumulator to reduce on-chain storage overhead.
- 2) Using one-way hash chains and certificateless post-quantum cryptography, we assign multiple unlinkable pseudonymous identities and corresponding key pairs to each device. This protects privacy and enables identity recovery for traceability.
- 3) We provide formal correctness analysis using BAN logic and prove that the protocol satisfies essential security properties. Our performance evaluation demonstrates that the scheme maintains low computational and communication overhead on resource-constrained devices.

Section II introduces mathematical foundations, system architecture, and security model. Section III details the authentication protocol. Section IV provides correctness and security analysis. Section V evaluates performance, and Section VI concludes the paper.

II. TRACEABLE ANONYMOUS AUTHENTICATION FRAMEWORK

This chapter presents the related mathematical assumptions, provides an overview of the system framework, summarizes the security requirements for cross-domain authentication, and introduces the security model.

A. Mathematical Assumptions

Elliptic Curve Discrete Logarithm Problem (ECDLP): Given two points K and $Q = xK$ on an elliptic curve, where $x \in \mathbb{Z}_l^*$ and l is a large prime number, it is computationally infeasible to determine the scalar x from K and L .

Elliptic Curve Computational Diffie-Hellman Problem (ECDHP): Given points K , $Q_1 = aK$, and $Q_2 = bK$

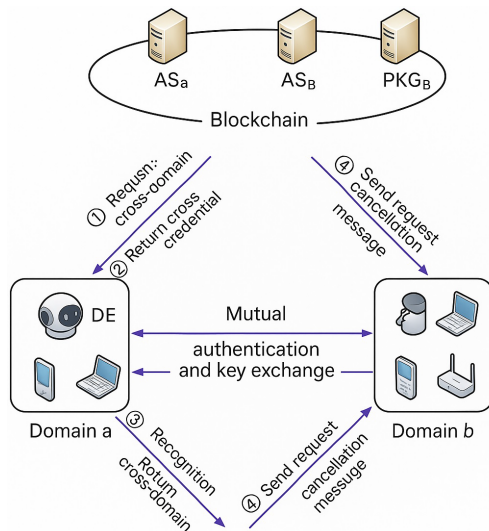


Fig. 1. System architecture showing cross-domain mutual authentication and key exchange based on blockchain.

on an elliptic curve, where $a, b \in \mathbb{Z}_l^*$, it is computationally hard to compute abK .

Strong Diffie-Hellman (K -SDH) Assumption: Let W be a cyclic group of prime order k , with generator $g \in W$, and $a \in \mathbb{Z}_k^*$. Given a $(l+1)$ -tuple $(w, l^a, l^{a^2}, \dots, l^{a^l})$, it is hard to compute a pair $(x, l^{1/(x+a)}) \in \mathbb{Z}_k \times W$.

B. System Framework

The system framework, as illustrated in Figure 1, consists of three main entities: IoT Device (DE), Private Key Generator (PK), and Authentication Server (AS).

- 1) **IoT Device (DE):** The IoT device possesses limited computational and storage capabilities. It cannot perform complex operations but can store pseudonymous identities and execute lightweight cryptographic computations.
- 2) **Private Key Generator (PK):** The PK is assumed to be fully trusted, with robust computing power and storage capacity. It is responsible for registering DEs and ASs within a domain, generating pseudonymous identities for DEs, issuing cross-domain credentials, and tracing malicious devices. As a consortium blockchain node, the PK publishes domain-related information on the blockchain.
- 3) **Authentication Server (AS):** The AS operates as a blockchain node. It queries the published domain information and assists the PK in completing device authentication, thereby reducing the computational burden on the PK.

C. Security Requirements

Security and privacy are the most critical factors in IoT. Due to the open, heterogeneous, and dynamic nature of IoT, along with the limited computational and storage capabilities of terminal devices [26], IoT systems are highly susceptible to various network attacks. Lattice-based certificateless cryptography's parameter selection process balances security against lattice attacks with performance and key sizes by selecting the right moduli, dimensions, and error distributions.

The post-quantum and collision resistance of hash functions are taken into consideration when choosing them. Larger parameters provide stronger security but slower calculation and larger key/signature sizes, requiring careful optimization for particular applications. Performance trade-offs result from the complexity of the underlying lattice issues. Based on the structure of IoT and previous studies [15], [16], [17], [18], the security requirements for cross-domain authentication and key agreement protocols in IoT scenarios can be summarized as follows:

- 1) **Mutual Authentication:** Both communicating parties must authenticate each other's identity before data transmission to establish trust.
- 2) **Anonymity:** To protect device privacy, no entity should be able to derive the real identity of a device from intercepted messages.
- 3) **Unlinkability:** An attacker should not be able to link multiple sessions from the same device. Anonymity, unlinkability, and untraceability are distinct but related concepts concerning privacy. Anonymity conceals the identity of the user, making it difficult to know who is performing an action. Unlinkability prevents actions from being connected to one another, even if the user is known.
- 4) **Traceability:** When malicious behavior occurs, regulatory mechanisms must be able to reveal the real identity of the device.
- 5) **Resistance to Common Attacks:** The protocol should resist common attacks such as replay attacks, impersonation attacks, and man-in-the-middle attacks.

D. Security Model

This section introduces a security model suitable for the proposed scheme, based on references [27], [28], [29]. Private Key Generator (PK) IoT systems are extremely vulnerable to a variety of network assaults because of the open, diverse, and dynamic nature of the Internet of Things as well as the constrained processing and storage power of terminal devices. BAN logic is used to verify correctness, and formal security analysis verifies that the scheme is resistant to typical attacks, including those that could be launched in a post-quantum setting. Instead of defining a specific collision-resistant structure, whether classical or post-quantum, BAN logic proofs in their original form rely on the characteristics of an ideal hash function. In some situations, involving digital signatures, this abstraction results in a "unsound" inference rule, which has been recognized as a flaw. Utilizing BAN logic, perform a formal correctness analysis and demonstrate that the protocol satisfies key security requirements. According to performance evaluation, the technique is appropriate for restricted IoT contexts since it achieves low computational and communication overhead. The model involves three types of participants: the Private Key Generator (PK), the Authentication Server (AS), and the IoT Device (DE), denoted as Π_{PK}^i , Π_{AS}^j , and Π_{DE}^k for their i -th, j -th, and k -th instances, respectively. The adversary $\mathcal{A}$ interacts with participants through various queries to compromise the security of the protocol. These queries are defined as follows:

- $\text{Execute}(\Pi_{PK}^f, \Pi_{AS}^j, \Pi_{DE}^t)$: This simulates a passive attack. $\mathcal{A}$ observes and obtains the messages exchanged among the participants.
- $\text{Send}(\Pi_{PK}^i, \Pi_{AS}^i, \Pi_{DE}^i, m)$: This models active attacks. $\mathcal{A}$ can impersonate a participant (e.g., Π_{DE}^i) and send message m to Π_{PK}^f or Π_{AS} , and receive the corresponding response if the message is valid.
- $\text{Hash}(x, h(x))$: A hash oracle that maintains a list of hash input-output pairs $(x, h(x))$. When $\mathcal{A}$ queries a value x , if x exists in the list, the corresponding $h(x)$ is returned. Otherwise, a random value $k \in \mathbb{Z}_l^*$ is generated, returned to $\mathcal{A}$, and stored in the list.
- $\text{Reveal}(\Pi_{DE}^b)$: This simulates a known-key security query, where the session key established between devices is revealed to $\mathcal{A}$. In Post-Quantum Traceable Anonymous Cross-Domain Authentication for IoT (a system), forward secrecy is maintained by making sure each session uses a unique key derived from a shared main secret or new random numbers, so that compromising an old session key does not compromise past or future data, even if a pseudonym is compromised. Session keys are refreshed at the end of each authentication session or by creating new sessions.
- $\text{Test}(\Pi_{DE}^b)$: This models the semantic security of the session key. Π_{DE}^b randomly selects a bit $b \in \{0, 1\}$. If $b = 1$, the real session key is returned to $\mathcal{A}$; otherwise, a random key of equal length is returned. $\mathcal{A}$ must guess whether the returned key is real or random by outputting a guess b' . If $b' = b$, then $\mathcal{A}$ wins the game and breaks the protocol's security.

Let Succ denote the event that $\mathcal{A}$ wins the game. The advantage of adversary $\mathcal{A}$ in breaking the protocol K is:

$$\text{Adv}_{\mathcal{A}}^{\text{AKA}} = |2 \cdot \text{Pr}[\text{Succ}] - 1| = |2 \cdot \text{Pr}[b' = b] - 1|. \quad (1)$$

Definition 1 (Semantic Security, AKA-Secure): Protocol K is said to be AKA-secure if for any probabilistic polynomial-time adversary $\mathcal{A}$, the advantage $\text{Adv}_{\mathcal{A}}^{\text{AKA}}$ is negligible.

III. CROSS-DOMAIN AUTHENTICATION PROTOCOL

A. Protocol Overview

This paper proposes a traceable and anonymous cross-domain authentication scheme based on blockchain technology, implemented using Hyperledger Fabric v1.4 with the Practical Byzantine Fault Tolerance (PBFT) consensus algorithm. The Practical Byzantine Fault Tolerance (PBFT) consensus method was not natively available as a standard ordering service option in Hyperledger Fabric v1.4, despite the fact that it supported pluggable consensus mechanisms. Although Hyperledger Fabric acknowledged and investigated the idea of PBFT and its advantages for Byzantine fault tolerance, v1.4 lacked a native, production-ready PBFT ordering service. During that time, most PBFT implementations in Fabric were either research projects or specialized integrations. The protocol comprises four phases:

- 1) System Initialization: Each domain's PK generates its domain-specific system parameters and broadcasts

TABLE I
SYMBOLS AND THEIR DESCRIPTIONS

Symbol	Description
$\mathcal{Z}_1 / \mathcal{Z}_2$	Zone 1 / Zone 2
ZID_x	Identifier of zone $\mathcal{Z}_x$
$\text{UID}_{m,n}$	Temporary identity of node $N_{m,n}$
K_{main}	Main key held by the trust authority
K_{TA}	Public key of the trusted authority
K_{prauth}	Private key of the authentication server
K_{pbauth}	Public key of the authentication server
r_1, r_2	Session-specific random values
(κ_1, κ_b)	Private key components of node $\text{UID}_{m,n}$
K_n	Public key corresponding to node $\text{UID}_{m,n}$

the transaction containing these parameters to the blockchain, laying the groundwork for future cross-domain authentication.

- 2) Registration: This phase includes AS registration and DE registration. To achieve traceable anonymous authentication, multiple unlinkable pseudonyms are generated for each device. During cross-domain authentication, devices use pseudonyms, which can later be traced back to their real identity by the PK if needed. To enhance privacy, certificateless cryptography is employed to generate public-private key pairs, eliminating the risk of private key leakage due to PK key escrow. To avoid high communication and storage overhead from uploading all pseudonymous public keys to the blockchain, the scheme uses a dynamic accumulator to store only the accumulated value of each domain on-chain, thus maintaining authentication functionality while reducing overhead.
- 3) Identity Revocation and Update: Each used pseudonym is revoked to ensure unlinkability of session messages.
- 4) Cross-Domain Authentication and Key Agreement: Before cross-domain communication, device $DE_{U_1}^v$ requests a cross-domain credential from its domain PK_1 . During authentication, the credential is presented to the external device $DE_{i/2}^v$. Due to limited storage on IoT devices, they cannot store blockchain data directly. Hence, $DE_{i/2}^v$ forwards the credential to AS_b for verification. AS_b verifies the credential against domain information stored on the blockchain and returns the authentication result to $DE_{i/2}^v$.
- 5) Parameter settings: According to the parameter settings, we assume that the hash outputs are 32 bytes, the true device identity is 32 bytes, the pseudonymous identity is 64 bytes, the domain identifier and version number are 4 bytes, and random numbers and seed values are 32 bytes. By creating its domain-specific system parameters and broadcasting the transaction with these parameters to the blockchain, each domain's PK establishes the foundation for upcoming cross-domain authentication.

Table II summarizes the key symbols used throughout the paper.

B. System Initialization

Taking domain $\mathcal{Z}_1$ as an example, PK_0 selects a security parameter l and generates the system public parameters:

$$(k_1, l, W_1, W_T, W, e_1, w_1, pks_{ac}, K, K_{am}, Y_1, Y_2, Y_3) \quad (2)$$

TABLE II
SYMBOLS AND THEIR NOTATIONS

Symbol	Description
Z_1/Z_2	Domain 1 / Domain 2
ZID_x	Domain identifier of domain i
$UID_{U,j}$	j -th pseudonym of device $DE_{i,j}$
K_{main}	Main secret key of PK_i
K_{AD_i}	Public key of PK_i
sk_{sAS}	Secret key of authentication server AS_i
U_{AS}^i	Public key of authentication server AS_i
SD_{v1}/SD_{v2}	Randomly selected seeds
(x_{1j}, z_{U1}^j)	Private key pair corresponding to $UID_{U1,j}$
K_{U1}	Public key corresponding to $UID_{U1,j}$

where:

- W_1, W_T, W are cyclic groups of prime order k_1 .
- $e_1 : W_1 \times W_1 \rightarrow W_T$ is a bilinear pairing.
- w_1 and K are generators of W_1 and W , respectively.
- l and k_1 are large prime numbers.
- $pk_{sac} = (w, l^{\Lambda_1}, \dots, l^{\Lambda_n}, u)$ and $sk_{sac} = \Lambda_0 \in \mathbb{Z}_{p_0}^*$ denote the public and private keys of the dynamic accumulator.
- The main secret key of the domain is $msk_1 = s_1 \in \mathbb{Z}_l^*$, and the corresponding public key is $K_{ps} = s_1 K$.
- $Y_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_l^*$, $Y_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_l^*$, and $Y_3 : \{0, 1\}^* \times W \rightarrow \mathbb{Z}_l^*$ are secure hash functions modeled as random oracles.

PK_1 then publishes an accumulator transaction:

$$ATX = (DIZ_1, v, k_1, q, W_1, W_T, W, e_1, w_1, K, U_{AS}, pk_{sAS}, \Omega, \text{sig}_{s_1}) \quad (3)$$

where:

- DIZ_1 is the domain identifier.
- U_{AS} is the public key of the authentication server.
- Ω is the initial accumulator value (initially empty).
- v is the domain information version number.
- sig_{s_1} is the digital signature of PK_1 over the transaction.

During this process, the blockchain serves as a decentralized ledger that stores essential authentication data such as the accumulator value and version number for each domain. Due to its decentralized and tamper-proof nature, the blockchain provides a reliable source of identity verification information during cross-domain authentication. A sophisticated attack technique known as “cross-domain credential leakage” allows illegal access to an apparently unrelated domain by using compromised credentials from one system or environment. Rather than “breaking in,” adversaries “log in” by taking advantage of the trust relationships and security flaws between cloud systems, identity providers, and interconnected endpoints. Password and username combinations that were made public in previous company breaches are automatically tested by attackers. This “credential stuffing” technique can result in numerous account compromises by taking advantage of the widespread password reuse practice. Systems such as Apache Spark synchronize accumulator changes across domains by first delivering task-specific updates to worker nodes (domains), and then aggregating these individual updates on the driver program when an action initiates the aggregation process. By using a dynamic accumulator to store only the total value of each domain on-chain, the technique preserves authentication

Algorithm 1 Registration Phase

Require: IoT Device D_i , Private Key Generator (PKG), System Parameters $params$

Ensure: Device obtains pseudonyms, keys, and registration proof

- 1: **// Step 1: Device generates identity**
- 2: D_i selects a unique identifier ID_i
- 3: D_i sends registration request $\{ID_i\}$ to PKG over a secure channel
- 4: **// Step 2: PKG generates pseudonyms and keys**
- 5: PKG computes a set of unlinkable pseudonyms $\{PID_i^1, PID_i^2, \dots, PID_i^n\}$
- 6: For each PID_i^j , PKG generates corresponding key pair (sk_i^j, pk_i^j) using post-quantum signature scheme (e.g., XMSS)
- 7: PKG stores mapping $\langle ID_i, \{PID_i^j\}, \{pk_i^j\} \rangle$ in a secure traceable log
- 8: **// Step 3: Credential issuance**
- 9: PKG computes credential $Cred_i$ using accumulator or blockchain anchor
- 10: PKG sends $\{PID_i^j, sk_i^j, Cred_i\}$ to D_i securely
- 11: **// Step 4: Device stores credentials**
- 12: D_i stores $\{PID_i^j, sk_i^j, Cred_i\}$ locally for future authentication

functionality while lowering communication and storage overhead that would result from uploading all pseudonymous public keys to the blockchain.

C. Registration Phase

The registration phase includes secure initialization of cryptographic credentials for both authentication servers (AS) and IoT devices (DE), enabling future anonymous yet traceable authentication across domains. Digital signatures using cryptographic primitives like XMSS and CRYSTALS-Dilithium provide protection from dangers posed by quantum computing. Researchers define the precise primitives, algorithms, and parameters utilized in their implementations to guarantee reproducibility in cryptographic systems. RYSTALS-Dilithium is a lattice-based digital signature technique that was recently standardized by NIST as Module Lattice-based Digital Signature Algorithm (ML-DSA). Because of its relative compactness, security, and performance, it is a key choice in the post-quantum cryptography standardization process. In order to improve Post-Quantum Traceable Anonymous Cross-Domain Authentication for IoT, the phrase refers to a planned or current implementation that makes use of hardware acceleration characteristics included in contemporary IoT chipsets. This makes use of specialized hardware on IoT devices to accelerate cryptographic processes, which are essential for a blockchain-based system that must offer cross-domain, anonymous, and secure authentication for IoT devices, especially in the face of growing threats from quantum computing. Researchers must empirically test a scheme to measure its memory usage during authentication and key exchange in order to confirm its applicability for limited IoT devices. They must then submit the results alongside

measurements from baseline schemes that are already in place for direct comparison. This comparison study offers verifiable proof of the scheme's computational resource efficiency, indicating its applicability in settings with severely constrained memory and processor capacity.

1) *Authentication Server Registration*: An authentication server (e.g., AS_0) securely registers with the private key generator (PK_1) by submitting its real identity ID_{AS} . The PKG generates a secret-public key pair using random values and a hash-based challenge-response mechanism. The computed public key U_{AS} and corresponding secret are returned to AS_0 , which verifies their correctness using a public verification equation. A secure and portable certificateless encryption method... Digital signature schemes vary greatly in complexity depending on the underlying algorithm. For example, RSA usually has high costs for key generation and signing because of its complex exponentiation, whereas ECC and lattice-based schemes offer speed and efficiency gains but have different computational trade-offs. Implementing robust security mechanisms on limited Internet of Things (IoT) devices inevitably results in trade-offs with performance parameters such as memory utilization, computing speed, and energy consumption. Developers must balance security strength against the device's and its operating environment's practical limits since standard security solutions are too resource-intensive. The plan uses quantum-resistant hash algorithms and lattice-based certificateless cryptography to guarantee resilience in the face of quantum attacks. Multiple unlinkable pseudonymous identities with corresponding post-quantum public-private key pairs are allocated to devices. Key and signature sizes, computational demands in resource-constrained situations (such as wireless networks), and the growing threat of quantum computers, which calls for the adoption of post-quantum cryptography (PQC), are additional variables that impact practicality. Upon successful verification, AS_0 stores its key pair for future protocol execution.

2) *IoT Device Registration*: A device $DE_{U_1}^{v_1}$ undergoes a multi-step registration process with PK_1 :

a) *Key seed generation*: The device selects $m+1$ random values as secrets and computes the corresponding public components. These are sent securely to PK_1 .

b) *Pseudonym and UID generation*: The PKG generates two seed values to derive m unlinkable pseudonyms and corresponding obfuscated user IDs ($UIDs$), ensuring anonymity and unlinkability.

c) *Key generation*: The PKG uses the received public components and internal secrets to generate long-term and pseudonym-based public-private key pairs. Each key is securely linked to the pseudonyms via hash-based derivations and signed to ensure traceability.

d) *Credential distribution and verification*: The PKG sends all pseudonyms, keys, and a derived accumulator element τ_{U_1} to the device. The device verifies the correctness of these credentials using pre-defined equations. On success, the device stores all necessary key materials locally.

e) *Accumulator update*: The PKG updates the global accumulator value Ω to reflect the new registration and publishes a blockchain transaction ATX' containing the updated state. A major element affecting the finality of blockchain

transactions is network latency; a high latency affects user experience and company confidence since it results in lengthier delays between the submission of a transaction and its irreversible recording on the blockchain. The speed at which a transaction becomes final is directly impacted by high latency in a blockchain network, which results in lengthier delays for broadcasting and confirming transactions. In blockchain, finality refers to the point at which a transaction is irrevocable and permanent. A transaction is locked in once it reaches finality and cannot be changed or reversed.

D. Identity Revocation and Update

To revoke pseudonym $UID_{U,j}$, PK_1 adds $uid_{1,j,1}$ to its local revocation list once a cross-domain credential is issued. Each pseudonym is single-use, ensuring unlinkability. The revocation process in a blockchain-based Post-Quantum Traceable Anonymous Cross-Domain Authentication system for the Internet of Things usually entails adding the user's identity (or the pseudonym that goes with it) to a distributed ledger or blockchain-based revocation list, which is then routinely verified by system nodes. Pseudonym revocation and frequent device registration are typical in systems that value user accountability and privacy. Frequent modifications are especially necessary in settings where there is a high danger of device compromise or where a user's location, identity, or activity habits could be tracked over time. To securely deploy and authenticate new devices at scale, automated registration is required. In order to acquire new cryptographic keys, devices might also need to re-register following power cycles or disconnections. By forcing the user to disconnect their existing connections and go through a thorough re-authentication process in order to recover access, this action instantly invalidates their credentials and ends all active sessions. Accountability is guaranteed, and malevolent devices are stopped from gaining unauthorized access after revocation.

When the number of remaining pseudonyms is less than a threshold L , the device may reapply for pseudonyms. The registration request is encrypted using PK_1 's public key. Upon decryption, PK_1 uses elliptic curve cryptography to encrypt new pseudonyms with X_{10} and sends them securely to the device.

E. Cross-Domain Authentication and Key Agreement

First, a device requests PK to issue a cross-domain credential—this can be done during idle time. Then, two devices from different domains authenticate each other and establish a session key.

As illustrated in Figure 2, the authentication process is demonstrated using device DE_{i1}^a from domain Z_1 and device DE_{U2}^b from domain Z_2 .

Step1: Request for Cross-Domain Credential

$$DE_{v_1} \rightarrow PK_1 : M_1 = \{UID_{v_1,j}, DID_{Z_2}, N_1, sig_{x_{16}}\} \quad (4)$$

DE_{v_1} , using the pseudonym identity $UID_{v_1,j}$, requests cross-domain authorization from PK_σ . DID_σ is the destination domain identifier, N_1 is a random nonce, and sig_{x_0} is the signature generated using the device's long-term private key $(x_{10}, z_{j1}^{\theta_1})$ (see Step3 for details).

Algorithm 2 Cross-Domain One-Way Authentication Process

Require: Device D_a in Domain a , Authentication Server AS_b in Domain b , Public parameters $params$

Ensure: Successful authentication of D_a to AS_b

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1: // Step 1: Device  $D_a$  initiates request
2:  $D_a$  selects pseudonym  $PID_a^i$ , private key  $sk_a^i$ , and generates random nonce  $r_a$ 
3:  $D_a$  computes signature  $\sigma_a = \text{Sign}_{sk_a^i}(PID_a^i \parallel r_a \parallel T_a)$ 
4:  $D_a$  sends message  $M_1 = \{PID_a^i, r_a, T_a, \sigma_a\}$  to  $AS_b$ 
5: // Step 2: Authentication Server  $AS_b$  verifies request
6:  $AS_b$  checks timestamp  $T_a$  for freshness
7:  $AS_b$  verifies  $\sigma_a$  using public key associated with  $PID_a^i$ 
8: if verification successful then
9:    $AS_b$  generates session key  $K_{ab}$  and random nonce  $r_b$ 
10:   $AS_b$  computes authentication token  $\tau_b = H(K_{ab} \parallel r_a \parallel r_b)$ 
11:   $AS_b$  sends  $M_2 = \{r_b, \tau_b\}$  to  $D_a$ 
12: else
13:   Reject authentication and abort protocol
14: end if
15: // Step 3: Device  $D_a$  verifies server response
16:  $D_a$  computes expected  $\tau'_b = H(K_{ab} \parallel r_a \parallel r_b)$ 
17: if  $\tau'_b = \tau_b$  then
18:   Authentication successful
19: else
20:   Abort protocol
21: end if

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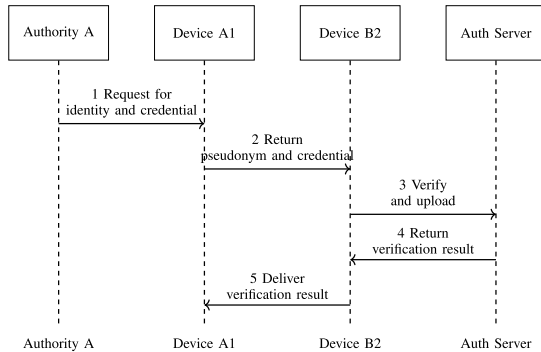


Fig. 2. Cross-Domain one-way authentication process using different devices and domains.

Step2: Generation and Issuance of Cross-Domain Credential

1) Upon receiving the request, PK_s checks if $UID_{U1,j}$ is in the revocation list. If not, it computes

$$ID'_{U1} = pid_{1,j-z} \oplus Y_3(s_1 pid_{1,j-1}, K_{sc+}) \quad (5)$$

It then retrieves the public key K_{v1}^4 from the local database and verifies the signature (see Step4 for verification method). If verification fails, the request is denied. Otherwise, it calculates the accumulator element τ_{v1} and generates the witness $w_{v1} = \Omega^{1/(A_0 a^{+*} - m)}$. Using DID_8 , it retrieves the domain public key from the blockchain and issues a credential:

$$v = \text{Enc}_{v_{ss}}(UID_{v1,j}, \tau_{v1}, w_{v1}, v) \quad (6)$$

where v is the version number of the domain information.

2) PK_1 returns the credential to DE_{U1}^d :

$$PK_1 \rightarrow DE_{v1} : M_2 = \{v, N_1, \text{sig}_{ast}\} \quad (7)$$

Step3: Send Authentication Request

$$DE_{v1}^v \rightarrow DE_{v2}^v : M_3 = \{UID_{v1,j}, Pu_{11}, DIZ_1, v, N_2, \text{sig}_{xu}\} \quad (8)$$

DE_{v1} verifies the integrity of M_2 using PK_1 's public key and ensures N_1 matches to prevent replay attacks. If valid, it sends M_3 using pseudonym $UID_{v1,j}$. The signature sig_{xu} is computed using the Schnorr signature scheme over elliptic curves:

$$\begin{aligned} w_1 &\in Z_v^* \text{ randomly chosen, } \tau_1 = w_1 K \\ \Gamma &= Y_1(\tau_1 \parallel M_3) \\ \zeta_1 &= w_1 + \Gamma(y_{1,j} x_{1j} + z \tilde{l}_1) \\ \text{sig}_{x1} &= (\tau_1, \zeta_1) \end{aligned} \quad (9)$$

Step4: Forward Cross-Domain Authentication Request

$$\begin{aligned} DE_{i/2}^\Omega &\rightarrow AS_b : M_4 = \{UID_{i/2,j}, \\ &K_{izj}^\Omega, DIZ_1, UID_{i1,j}, v, N_3, \text{sig}_{x1}\} \end{aligned} \quad (10)$$

$DE_{i/2}$ verifies the signature by computing:

$$\begin{aligned} y'_{1,j} &= Y_1(pid_{1,j-1} \parallel pid_{1,j-2} \parallel Put_1) \\ y'_{2,j} &= Y_2(pid_{1,j-1} \parallel pid_{1,j-2} \parallel K_{v1}^v) \\ \Gamma &= Y_1(\tau_1 \parallel M_3) \\ \chi &= y'_{1,j} \cdot K_{v1}^v + y'_{2,j} K_{ab,b} \\ \text{verify: } \zeta K &= \tau_1 + \Gamma \chi \end{aligned} \quad (11)$$

If verified, store N_2 , add N_3 , and send M_4 to AS_b .

Step5: Cross-Domain Authentication at AS_b

1) AS_b verifies the signature, decrypts v to extract $UID_{U1,j}$, τ_{U1} , w_{U1} , v . It checks consistency with M_4 , retrieves Ω and validates:

$$e(w_{v1}, w_1^{gen} w_1^\Lambda) = e(\Omega, w_1) \quad (12)$$

2) Return Result:

$$AS_b \rightarrow DE_{1/2}^b : M_5 = \{N_3, \text{success}, \text{sig}_{is}\} \quad (13)$$

Step6: Reverse Authentication

$DE_{i/2}$ checks the signature and nonce. If valid, it accepts the authentication. Reverse authentication from DE_{v1} to DE_{v2} follows a similar process, with nonce N_4 and included N_2 .

Step7: Session Key Agreement

After mutual authentication, both parties compute:

$$SE = w_1 \tau_2 = w_2 \tau_1 = w_1 w_2 P \quad (14)$$

Step8: Challenge-Response Completion

$$DE_{i/1}^v \rightarrow DE_{i/2}^v : M_6 = \{N_2, N_4, \text{HMAC}_{SE}\} \quad (15)$$

After verifying N_2 , both sides complete the session and delete the pseudonym.

IV. PERFORMANCE ANALYSIS

This section evaluates the proposed scheme through experiments from two aspects: computational overhead and communication overhead. The experiments were conducted on Windows 11 OS, with 16.0GB RAM and an AMD Ryzen™ 7 5800H @ 3.20GHz hardware platform.

TABLE III
COMPARISON OF SECURITY ATTRIBUTES

Security Attribute	BLMA [15]	CABC [17]	UBAKE [18]	Proposed Scheme
Mutual Authentication	Yes	Yes	Yes	Yes
Anonymity	Yes	Yes	Yes	Yes
Unlinkability	No	Yes	No	Yes
Traceability	No	No	Yes	Yes
Known-Key Security	Yes	Yes	Yes	Yes
Resistance to Replay Attacks	No	No	Yes	Yes
Resistance to Impersonation	No	Yes	Yes	Yes
Resistance to MITM Attacks	No	Yes	No	Yes

TABLE IV
STATS ON CRYPTOGRAPHY OPERATIONS

Entity	Computation
DE_{U1}^+	2CSM + 2GFMul + 2GFAdd + H
DE_{U2}^+	9CSM + 4CPA + 2GFMul + 2GFAdd + 7H
AS_b	5CSM + 2CPA + 3GFMul + 2GFAdd + GEXP + 2BP + De

A. Computational Overhead

To assess the computational cost, we analyze the operations involved in cross-domain authentication and key agreement. The computational cost on the device side is simulated and compared with other schemes. The symbols used are defined as follows:

- CSM/CPA: Scalar multiplication/addition on elliptic curves.
- GFMul/GFAdd: Multiplication/addition over finite field $\mathbb{Z}_k$.
- GSM/GEXP: Multiplication/exponentiation in a cyclic group W .
- H/En/De/BP: Hash operation/public key encryption/decryption/bilinear pairing.

1) *Theoretical Analysis*: Table IV lists the cryptographic operations performed by each entity during one-way authentication and key agreement. In the authentication process where $DE_{i/2}^v$ authenticates $DE_{i/1}^v$, the message is forwarded to AS_b for verification. Devices $DE_{i/1}^v$ and $DE_{i/2}^v$ do not execute complex bilinear pairings or exponentiation operations.

2) *Simulated Experiment*: We compare the computational overhead during cross-domain authentication and key agreement with schemes from [15], [17], and [18]. Since servers are assumed to have powerful computational resources, their overhead is not included in the comparison.

In our experimental setup, we use SHA-256 and the elliptic curve secp256r1. The measured average execution times of operations are:

- CSM: 0.0511 ms
- CPA: 0.007 ms
- GFMul: 0.0012 ms
- GFAdd: 0.0003 ms
- H: 0.0001 ms
- GEXP: 0.0298 ms

Table V presents the computation costs of each scheme on the device side. Existing blockchain-based schemes are

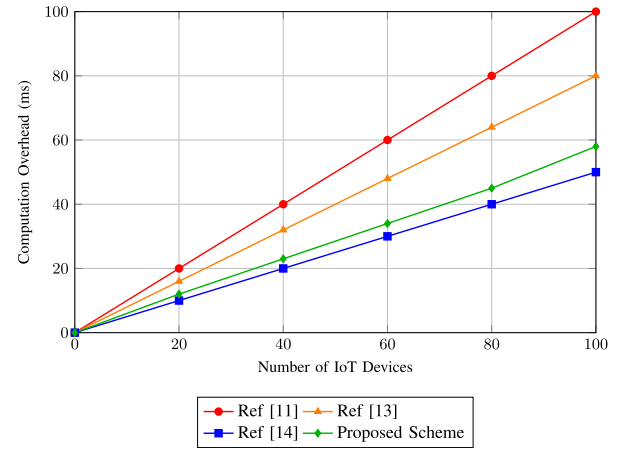


Fig. 3. Data from simulations demonstrating device computation overhead while key negotiation and cross-domain authentication.

typically classified as either device-to-device or device-to-server authentication. Our scheme and those from [15] and [17] fall into the former category, while [18] is the latter. Each system's quantum resistance must be increased by precisely outlining its underlying mathematical hardness assumptions and supplying continuous cryptanalysis results to back up its long-term security claims against both potential classical attacks and sophisticated quantum computers in order to guarantee the dependability of quantum-resistant cryptography. This method, which is used in the NIST PQC standardization process, entails choosing algorithms according to particular mathematical issues. The presumptive complexity of some mathematical problems that are believed to be unsolvable by both classical and quantum computers is the foundation of quantum-resistant cryptography techniques.

Device-to-device authentication must handle end-to-end security without assuming full blockchain data availability on constrained devices. Thus, such schemes usually include more signature operations, leading to higher overhead than device-to-server protocols. However, our scheme achieves lower computation cost than both [15] and [17]. As shown in Figure 3, when the number of devices is 200, our scheme reduces the computation cost by approximately 42.73% and 26.33% compared to [15] and [17], respectively.

B. Communication Overhead

This section compares the communication overhead incurred during cross-domain authentication and key agreement with other schemes. Based on the parameter settings, we assume the domain identifier and version number are 4 bytes, random numbers and seed values are 32 bytes, hash outputs are 32 bytes, real device identity is 32 bytes, and pseudonymous identity is 64 bytes. Elements in W_{a1} , W_{a2} , and $\mathbb{Z}_l^*$ are 32 bytes long, while elements in group W are 64 bytes long.

During cross-domain authentication and key agreement, device DE_{U1}^v first sends a cross-domain request M_3 to DE_{U2}^v , which incurs a 392-byte communication cost. Then, $DE_{i/2}^v$ sends a message M_4 containing the cross-domain credential to AS_b , consuming another 392 bytes. Finally, $DE_{0/2}^b$ receives a 128-byte authentication success message and a 96-byte

TABLE V
COMPARISON OF COMPUTATION OVERHEAD ON DEVICES DURING CROSS-DOMAIN AUTHENTICATION AND KEY NEGOTIATION

Scheme	Computation	Total Time (ms)
BLMA [15]	$18\text{CSM} + 6\text{CPA} + 6\text{GFMul} + 7\text{GFAdd} + 13H + \text{GEXP}$	0.9687
CABC [17]	$14\text{CSM} + 4\text{CPA} + 5\text{GFMul} + 4\text{GFAdd} + 16H + 2\text{GEXP}$	0.7900
UBAKE [18]	$8\text{CSM} + 3\text{CPA} + 2\text{GFMul} + 2\text{GFAdd} + 5H$	0.4175
Proposed Method	$11\text{CSM} + 4\text{CPA} + 4\text{GFMul} + 4\text{GFAdd} + 8H$	0.5749

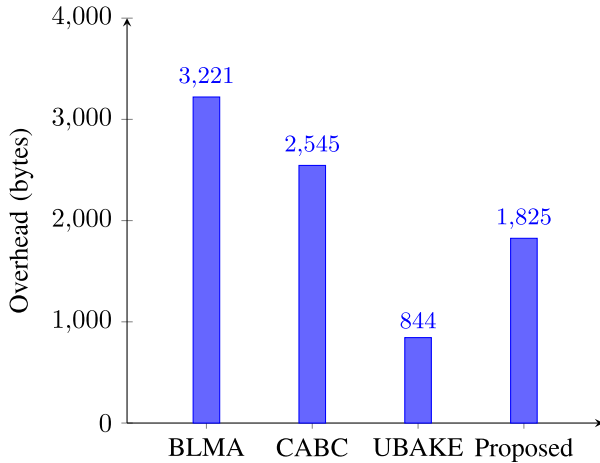


Fig. 4. Communication overhead comparison.

challenge-response message. Repeating the process for mutual authentication and key agreement, the total communication overhead is 1920 bytes. Analysis of particular PQC revocation algorithms in a blockchain-based IoT system is necessary for a thorough comparison. Nonetheless, popular state-of-the-art methods include off-chain revocation with on-chain anchors, which reduces blockchain storage and update costs but depends on a reliable intermediary component; revocation by updating device keys on the blockchain, which has high update costs but little storage; and using smart contracts to manage revocation lists, which balances storage and update efficiency. For IoT devices with limited resources, effective trade-offs between security, affordability, and scalability are essential because post-quantum techniques will inevitably have greater storage and processing costs because of larger key sizes and complicated algorithms.

Similarly, the communication overheads for the schemes in [15], [17], and [18] are 3221 bytes, 2545 bytes, and 844 bytes, respectively. Figure 4 shows the comparison. Although the scheme in [18] has the lowest communication overhead, its use of a single pseudonym per device poses a risk of linkability attacks. Even if the scheme generates a pseudonym per device using a pseudonym mechanism to achieve anonymity, an attacker can correlate the pseudonymized sessions using the message signatures and blockchain-stored data. Collecting and analyzing such session messages can eventually reveal the real identity of the device, thus undermining the anonymity and violating the privacy-preserving design goals. Decentralized identity management, in conjunction with strong key management systems and possibly privacy-preserving protocols, is a common approach for large-scale pseudonym management across millions of devices in various administrative domains. Decentralized or federated

systems that can issue credentials with embedded attributes are necessary for generation; secure, scalable infrastructure is used for distribution; and effective, privacy-aware mechanisms, such as MHTs, are used for revocation in order to broadcast certificate status without disclosing details about queries or queried targets. Pseudonyms can be used to link sessions and deduce user activity. However, strong anonymity can also protect bad actors. Privacy, responsibility, and traceability must all be guaranteed by a well-balanced design. Pseudonyms are often used in anonymous authentication systems. However, methods like those in have the problem of reusing pseudonyms, which permits privacy leakage and session linkage. Each device is assigned a number of unlinkable pseudonyms in order to accomplish traceable anonymous authentication.

To achieve anonymity and unlinkability in cross-domain device authentication, multiple pseudonyms are generated per device, and authentication information is uploaded to a consortium blockchain maintained by servers in each domain. The presumed trust boundaries between domains are crucial and need careful thought in a blockchain-based system, especially when rogue authentication servers are present. The model must take into consideration the possibility of a hostile or compromised authentication server within a domain, even though it is typically assumed that a domain's internal authentication servers are honest and carry out their duties effectively. This implies that for cross-domain authentication, the system cannot rely exclusively on the reliability of a single authentication server. However, IoT devices have limited storage capacity and cannot act as blockchain nodes to store all authentication data. As a result, they rely on servers to complete authentication, which inevitably increases communication overhead.

Compared with other device-to-device cross-domain authentication schemes, such as those in [15] and [17], our proposed scheme has the lowest communication overhead, indicating that it maintains a reasonable and acceptable communication cost.

V. CONCLUSION

To mitigate privacy leakage and enable traceability in cross-domain IoT authentication, this paper proposes a blockchain-based traceable anonymous authentication scheme designed with post-quantum security considerations. While the accumulator structure enables effective and anonymous device identity verification within the network, guaranteeing privacy and security, the blockchain offers a tamper-proof ledger for storing authentication data and maintaining trust. The scheme integrates one-way hash chains and certificate-less cryptography to enable each device to generate multiple unlinkable pseudonyms. This protects identity privacy while allowing the key generation center (KGC) to recover the real identity of malicious devices, ensuring accountability.

Correctness is verified using BAN logic, and formal security analysis confirms the scheme's resistance to common attacks, including those feasible in a post-quantum context. Performance evaluation shows that the scheme achieves low computational and communication overhead, making it suitable for constrained IoT environments. By balancing anonymity and traceability, the proposed approach supports secure multi-domain cooperation and lays the groundwork for robust malicious device detection and accountability. Experimental results demonstrate that our scheme outperforms existing methods in both computational efficiency and communication cost during cross-domain authentication. Specifically, authentication delay and key generation time remain minimal, even when scaling to a large number of domains and devices. The fine-grained control of the access rights and capabilities of various device subjects is a crucial assurance for the privacy protection of IoT data and system users, especially as the number of IoT devices and network scale grow. As a result, future research will focus more on more granular access control for safe data exchange across various IoT devices or networks. Moreover, the use of dynamic accumulators significantly reduces on-chain storage requirements, making the scheme practical for blockchain-based IoT deployments. In future work, we aim to extend this framework by incorporating a reputation-based trust evaluation mechanism to automatically identify and isolate malicious devices, further strengthening decentralized trust and security in post-quantum IoT networks.

REFERENCES

- [1] E. L. C. Macedo et al., "On the security aspects of Internet of Things: A systematic literature review," *J. Commun. Netw.*, vol. 21, no. 5, pp. 444–457, 2019.
- [2] S. Berlatto, U. Morelli, R. Carbone, and S. Ranise, "A secure and quality of service-aware solution for the end-to-end protection of IoT applications," *J. Netw. Comput. Appl.*, vol. 242, Oct. 2025, Art. no. 104246.
- [3] H. Zhong, L. Chen, J. Cui, J. Zhang, I. Bolodurina, and L. Liu, "Secure and lightweight conditional privacy-preserving authentication for fog-based vehicular ad hoc networks," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8485–8497, Jun. 2022.
- [4] K. B. A. Kumar, L. S. Mohith, K. Jain, P. Krishnan, N. Venkatachalam, and R. Buyya, "Post-quantum cryptography-based multimedia encryption communication scheme in IoT consumer electronics," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4995–5006, May 2025.
- [5] N. K. Singh, S. Patni, S. Lim, and J. Lee, "Federated learning-based secure computing mechanism for consumer Internet of Vehicles-based transportation cyber-physical systems," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4867–4882, May 2025.
- [6] S. Zeng, B. Cao, Y. Sun, C. Sun, Z. Wan, and M. Peng, "Blockchain-assisted cross-domain data sharing in industrial IoT," *IEEE Internet Things J.*, vol. 11, no. 16, pp. 26778–26792, Aug. 2024.
- [7] M. Shahbaz Khan et al., "Chaotic quantum encryption to secure image data in post quantum consumer technology," *IEEE Trans. Consum. Electron.*, vol. 70, no. 4, pp. 7087–7101, Nov. 2024.
- [8] C. Geng, Y. Zhang, X. Xu, Y. Yao, C. Lu, and Z. Zhi, "Blockchain-based identity authentication and data interaction scheme for industrial Internet of Things," *Comput. Electr. Eng.*, vol. 123, Apr. 2025, Art. no. 110143.
- [9] J. Hu, D. Sun, W. Lu, J. Dong, and H. Wu, "Blockchain-enabled distributed authentication mechanism for industrial device access," *IEEE Trans. Ind. Inform.*, vol. 21, no. 4, pp. 2819–2828, Apr. 2025.
- [10] M. K. Hasan, Z. Weichen, N. Safie, F. R. A. Ahmed, and T. M. Ghazal, "A survey on key agreement and authentication protocol for Internet of Things application," *IEEE Access*, vol. 12, pp. 61642–61666, 2024.
- [11] U. K. Lilhore, S. Simaiya, S. Dalal, A. Alshuhail, and A. Almusharraf, "A post-quantum hybrid encryption framework for securing biometric data in consumer electronics," *IEEE Trans. Consum. Electron.*, vol. 71, no. 3, pp. 8289–8297, Aug. 2025.
- [12] Z. Li, X. Liu, N. Zhang, and Z. Wei, "Blockchain-based certificateless cross-domain authentication scheme in the industrial Internet of Things," *Comput., Mater. Continua*, vol. 80, no. 3, pp. 3835–3854, 2024.
- [13] L. Xue, H. Huang, F. Xiao, and W. Wang, "A cross-domain authentication scheme based on cooperative blockchains functioning with revocation for medical consortiums," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 3, pp. 2409–2420, Sep. 2022.
- [14] S. Zhang, Z. Yan, W. Liang, K.-C. Li, and B. Di Martino, "BCAE: A blockchain-based cross domain authentication scheme for edge computing," *IEEE Internet Things J.*, vol. 11, no. 13, pp. 24035–24048, Jul. 2024.
- [15] F. Wang, J. Cui, Q. Zhang, D. He, C. Gu, and H. Zhong, "Blockchain-based lightweight message authentication for edge-assisted cross-domain industrial Internet of Things," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 4, pp. 1587–1604, Jul. 2024.
- [16] M. Zeng, J. Cui, Q. Zhang, H. Zhong, and D. He, "Efficient revocable cross-domain anonymous authentication scheme for IIoT," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 996–1010, 2025.
- [17] L. Feng, F. Qiu, K. Hu, B. Yu, J. Lin, and S. Yao, "CABC: A cross-domain authentication method combining blockchain with certificateless signature for IIoT," *Future Gener. Comput. Syst.*, vol. 158, pp. 516–529, Sep. 2024.
- [18] A. Shahidinejad, J. Abawajy, and S. Huda, "Untraceable blockchain-assisted authentication and key exchange in medical consortiums," *J. Syst. Archit.*, vol. 151, Jun. 2024, Art. no. 103143.
- [19] F. Zhao, H. Ding, C. Li, Z. Su, G. Liang, and C. Yang, "A blockchain-based efficient cross-domain authentication scheme for Internet of Vehicles," *Comput., Mater. Continua*, vol. 80, no. 1, pp. 567–585, 2024.
- [20] X. Chen, Q. Cheng, W. Yang, and X. Luo, "A novel blockchain-based anonymous roaming authentication scheme for VANET," *J. Inf. Secur. Appl.*, vol. 88, Feb. 2025, Art. no. 103922.
- [21] A. Sharma and S. Rani, "Post-quantum cryptography (PQC) for IoT-consumer electronics devices integrated with deep learning," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4925–4933, May 2025.
- [22] L. Malina, L. Popelova, P. Dzurenda, J. Hajny, and Z. Martinasek, "On feasibility of post-quantum cryptography on small devices," *IFAC-PapersOnLine*, vol. 51, no. 6, pp. 462–467, 2018, doi: [10.1016/j.ifacol.2018.07.104](https://doi.org/10.1016/j.ifacol.2018.07.104).
- [23] U. Banerjee, A. Pathak, and A. P. Chandrakasan, "An energy-efficient configurable lattice cryptography processor for the quantum-secure Internet of Things," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2019, pp. 46–48, doi: [10.1109/ISSCC.2019.8662528](https://doi.org/10.1109/ISSCC.2019.8662528).
- [24] T. Fritzmann and J. Sepulveda, "Efficient and flexible low-power NTT for lattice-based cryptography," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, May 2019, pp. 141–150, doi: [10.1109/HST.2019.8741027](https://doi.org/10.1109/HST.2019.8741027).
- [25] D. S. Gupta, S. H. Islam, M. S. Obaidat, A. Karati, and B. Sadoun, "LAAC: Lightweight lattice-based authentication and access control protocol for e-health systems in IoT environments," *IEEE Syst. J.*, vol. 15, no. 3, pp. 3620–3627, Sep. 2021, doi: [10.1109/JSYST.2020.3016065](https://doi.org/10.1109/JSYST.2020.3016065).
- [26] X. Wu and J. Liang, "A blockchain-based trust management method for Internet of Things," *Pervas. Mobile Comput.*, vol. 72, Apr. 2021, Art. no. 101330.
- [27] M. Shariq, N. Jamil, G. S. Rawat, S. A. Chaudhry, M. Masud, and A. Cangelosi, "An anonymous and privacy-preserving lightweight authentication protocol for secure communication in UAV-assisted IoAV networks," *Comput. Commun.*, vol. 239, Jul. 2025, Art. no. 108192.
- [28] D. Chattaraj, B. Bera, A. K. Das, S. Saha, P. Lorenz, and Y. Park, "Block-CLAP: Blockchain-assisted certificateless key agreement protocol for Internet of Vehicles in smart transportation," *IEEE Trans. Veh. Technol.*, vol. 70, no. 8, pp. 8092–8107, Aug. 2021.
- [29] X. Li, D. Yang, X. Zeng, B. Chen, and Y. Zhang, "Comments on 'Provably secure dynamic id-based anonymous two-factor authenticated key exchange protocol with extended security model,'" *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 12, pp. 3344–3345, Dec. 2019.